Transducer Belief Decomposition

Recovering input–HMM and output–transducer belief geometry in a transformer

MARS 4.0 – Simplexity

July 22, 2026

One-slide summary

- **Question (aim §10.4):** when a transformer is trained on the *joint* input/output stream of an ε-transducer, does its residual stream linearly encode *both* the input–HMM belief and the output–transducer belief, as *separable* quantities?
- **Construction:** autonomous input HMM $\hat{T}^{(y)}$ drives a transducer $\hat{T}^{(x|y)}$; observable is the composite token (y, x); joint kernel $= \mathrm{kron}\big(\hat{T}^{(y)}, \hat{T}^{(x|y)}\big)$.
- **Why a transducer (not the old composite-Mess3):** the old experiment had one observable stream and a *hidden* driver – a latent-disaggregation problem, not an input→output decomposition. There was no input symbol to build an input belief from.
- **Headline result:** the belief is linearly present ($R^2 \approx 0.99$ on SNS and Fractal2); observing the intermediate token gives a *separable* belief, hiding it *entangles* it and erases **95–98%** of the input belief while the transducer belief is kept – and the effect vanishes on an unpredictable-process control.
- **Deliverables:** library kernels + tests, two torch/numpy notebooks (feed-forward + coarse-graining, and feedback), docs & slides. Cross-checked against `explorer.html`.

Background: belief states / MSP (recap)

- HMM observer maintains a **belief (mixed) state** – posterior over hidden states given the past, updated by Bayes:

$$\langle \eta_{t+1} | \propto \langle \eta_t | \ T^{(x_t)}, \qquad \text{renormalize.}$$

- The reachable set of belief states is the **Mixed-State Presentation (MSP)**.
- **Linearity hypothesis (Shai et al. 2024):** for a transformer trained on next-token prediction, $\exists$ linear $\hat{M}$ with $\hat{M} f(x_{0:L}) \approx |\rho(x_{0:L})\rangle$ – the MSP is *linearly embedded* in the residual stream.
- Unifilar processes (golden mean, even) $\Rightarrow$ *finite* MSP. Nonunifilar processes break this.

The SNS: why it is the right probe target

Simple Nonunifilar Source (2 states, alphabet $\{0, 1\}$):

$$T^{(0)} = \begin{pmatrix} p & 1-p \\ 0 & p \end{pmatrix}, \qquad T^{(1)} = \begin{pmatrix} 0 & 0 \\ 1-p & 0 \end{pmatrix}.$$

- **Nonunifilar:** a symbol does not pin down the next state $\Rightarrow$ uncertainty accumulates.
- **Synchronizing symbol:** emitting a 1 resets the belief to certainty.
- $\Rightarrow$ **countably-infinite** belief set $\{\eta_0, \eta_1, \eta_2, \dots\}$: one point per run-length of 0s since the last 1, marching along the 1-simplex toward a limit; each 1 jumps back to η_0.

process	unifilar?	MSP
golden mean / even	yes	finite (a few points)
SNS	no, synchronizing	**countably infinite (an arc)**
Mess3	no, non-synchronizing	uncountable (fractal)

From generator to transducer

- A **generator** (coin, golden mean, SNS) emits one stream autonomously.
- A **transducer** is a channel with memory: reads input y, emits output x, kernel $\hat{T}^{(x|y)}$ = "given input y and state, emit x and transition".

Feed-forward, fully observable (from `explorer.html`):

- input HMM $\hat{T}^{(y)}$ emits y (states R);
- transducer $\hat{T}^{(x|y)}$ reads y, emits x (states S);
- observer sees *both* $\Rightarrow$ composite token (y, x), index $k = y\, n_x + x$.

$$\boxed{T^{(y,x)}_{\text{joint}} \;=\; \mathrm{kron}\big(\hat{T}^{(y)},\ \hat{T}^{(x|y)}\big)} \quad \text{on the joint state } (R, S).$$

Summed over the composite token this is row-stochastic $\Rightarrow$ a valid HMM the transformer trains on with ordinary next-token prediction.

Three belief targets (marginals of the joint belief)

target	definition	Stage A (IID input)
input belief	marginalize out S ($\to R$)	trivial (single point)
transducer belief	marginalize out R ($\to S$)	countable SNS arc
joint belief	full (R, S)	= transducer belief

- We fit *separate* linear probes to each target (per layer + concatenated).
- Decomposition question: are input and transducer beliefs carried in *distinct* residual-stream directions?

Why old composite-Mess3 was *not* input→output

composite-Mess3 (prior work)

- one observable token stream;
- single hidden state (driver, transducer);
- the "driver" is a *latent* the net must infer;
- probes the 6D joint belief + its driver/transducer *marginals*.

transducer (this work)

- two observable streams y and x;
- explicit channel $\hat{T}^{(x|y)}$ with its own state;
- input is *observed*, so an input belief exists;
- probes input belief vs transducer belief as separable objects.

Old = correct hierarchical-HMM latent disaggregation. New = the actual transducer decomposition of aim §10.4.

The crux: does the transformer see the intermediate token?

Coarse-Graining Roots §556. With input HMM $\hat{T}^{(y)}$ (states R) driving transducer $\hat{T}^{(x|y)}$ (states S):

- **Both y and x observed:** $V^{(y,x)} = \hat{T}^{(x|y)} \otimes \hat{T}^{(y)}$. Beliefs are **tensor products** $\Rightarrow$ input & transducer beliefs *separable*, expressible in $(|R|-1) + (|S|-1)$ dims.
- **Coarse-grain intermediate y (only x observed):**

$$W^{(x)} = \sum_y \hat{T}^{(x|y)} \otimes \hat{T}^{(y)}$$

  the sum over hidden y **entangles** R and S $\Rightarrow$ beliefs fill the full $|R||S|-1$ simplex (the cascade tetrahedron).
- **Coarse-grain output x:** reduces to the input HMM's MSP.

Verified numerically: fully-observable beliefs **100% rank-1 (separable)**; coarse-grained **34%** (SNS$\to$SNS), **6%** (Fractal2$\to$Fractal2, b=0.85).

Reinterpreting the old composite-Mess3 result

- composite-Mess3 hid the driver (the intermediate variable) $\Rightarrow$ it was the **coarse-grained** regime W, not a clean input$\rightarrow$output decomposition.
- Hence: joint belief R^2 high (0.98) *but* driver marginal lags (0.94; 0.56 at L0).
- That lag is the **expected signature of entanglement**, not a model failure. The research question becomes: *how much information does the transducer erase?*

Three observation regimes (one `OBSERVE` switch)

`OBSERVE`	tokens	operator	predicted belief
`joint`	(y, x)	V	product (separable)
`output_only`	x only	W	entangled cascade
`input_only`	y only	$\sum_x V$	input HMM MSP

- Same (R, S) space in all three $\Rightarrow$ always probe joint / input / transducer beliefs.
- Processes: `iid`, `sns`, `fractal2`, `mess3` (input and transducer).
- **Erase-gap**: run `joint` vs `output_only` on the same process; compare marginal R^2.
- **Stage C**: feedback / partially observable (modified-operator MSP) – implemented as a separate notebook.

Notebook pipeline (end to end)

1. **Build + verify** joint kernel; stationary dist; **entropy rate computed in-notebook** (the CE-loss floor).
2. **Ground-truth manifolds** (step 1): BFS-enumerate the joint MSP, marginalize, plot input vs transducer beliefs *separately*.
3. **Train** a 4-layer transformer ($\sim$146k params, Shai A.6 arch, N_{ctx}=50) on the active-regime token stream (1M steps full; 100k on free-T4).
4. **Loss vs entropy floor** sanity plot.
5. **Linear probes** (step 3) for input / transducer / joint, per layer + concat, *with an untrained-model baseline for every target*.
6. **Controls:** sequence-level CV, shuffle, temporal split.
7. **Decisive figure:** predicted-vs-true transducer belief with the countable SNS arc as a rug.
8. **Save** weights / config / history / probes / controls.

Belief–activation alignment (no off-by-one)

- Belief is computed by Bayesian filtering along each sampled sequence; η_t = posterior *after* tokens $0{:}t$.
- Residual-stream activation at position t (causal) has also seen tokens $0{:}t$.
- $\Rightarrow \eta_t \leftrightarrow a_t$ aligned exactly (the common silent bug; checked).
- Probe: OLS with intercept, aggregate multi-output R^2 and MSE; concatenated probe uses all 4 layers (256D).

Success criteria & expected quirks

- **Geometry:** predicted transducer belief lands on the $y = x$ line and on the *countable SNS arc* (decisive figure); high transducer R^2.
- **Controls:** seq-CV $\approx 1\times$ full MSE (no overfitting); shuffle $\gg 1\times$ (genuine pairing); untrained baseline $\gg$ trained.
- **Prediction:** CE loss converges near the computed entropy-rate floor; report $d_\mu = \frac{1}{L}\sum_t \mathrm{KL}(P_t \| Q_t)$ (geometric *and* predictive axes).
- **Separability:** analytical fraction rank-1 (≈ 1 for `joint`, < 1 for `output_only`).

Expected, not bugs (Stage A only): joint $\equiv$ transducer rows (IID $\Rightarrow$ joint state $=$ transducer state); input-belief $R^2 =$ NaN (constant target). Both become meaningful in Stage B.

Results — belief recovery (fully observed) & prediction

All runs: free-T4 Colab, 100k steps; loss converges to the computed entropy-rate floor ($d_\mu \approx 10^{-3}$ nats).

- **IID → SNS (validation):** transducer $R^2 = 0.994$; residual-stream readout lands on the countable SNS arc; shuffle 168×, untrained 46×, seq-CV 1.00×.
- **SNS → SNS,** `joint`**:** input R^2=0.99, transducer 0.99, separability 1.00 (product manifold); shuffle 127×.
- **Fractal2 → Fractal2,** `joint`**:** input R^2=0.997, transducer 0.995, separability 1.00; shuffle 218×. Belief geometry cleanly linear on both processes.

Results — erasure under coarse-graining (the headline)

Cross-fit: how well the *x-only* model recovers the **fully-observable** belief (low R^2 = erased):

process	separability (joint→out)	FULL **input** R^2	FULL **transducer** R^2
SNS → SNS	1.00 → 0.34	0.05 (∼**95% erased**)	0.89 (retained)
Fractal2 → Fractal2	1.00 → 0.06	0.017 (∼**98% erased**)	0.92 (retained)

- Coarse-graining the intermediate token **erases the input belief but preserves the transducer belief**, on two independent processes; more entanglement (lower separability) ⇒ more input erasure.

Results — boundary control (effect is contingent on predictability)

- **Fractal2 (b=0.5) → Fractal2:** the joint process is maximally random (entropy rate $= \log 4$, the vocab bound), so next-token prediction is trivial.
- Consequence: the transformer has *no incentive* to build belief states — probe $R^2 \approx$ untrained (0.76 vs 0.73), shuffle only 4× (vs 130–290×).
- $\Rightarrow$ the belief-geometry phenomenon **requires a predictable process**; it is not an artifact. (Notebook now auto-flags entropy $\approx \log(\text{vocab})$.)

What was added to the codebase

Library (`transition_matrices.py`):

- `driven_transducer` – fully-observable joint kernel V;
- `coarse_grained_transducer` – intermediate-hidden operator $W^{(x)}$;
- `input_only_operator` – output-hidden operator (input HMM);
- `iid_sns_transducer`, `_sns_kernel`, `_fractal2_kernel`.

Tests: shape/stochasticity, input marginal, Kronecker structure, coarse-grained & input-only operators.

Notebooks (torch/numpy only): feed-forward + coarse-graining (`transducer_decomposition_experiment.ipynb`); feedback / partially-observable (`transducer_feedback_experiment.ipynb`).

Docs + slides + README.

Verification & rigor

- Joint kernel and belief trajectories cross-checked against the `explorer.html` `sampleDrivenTransducer` reference: **identical beliefs** (max diff 0.0).
- Confirmed SNS transducer belief set is **countable** (e.g. 133 distinct states by depth-12 BFS); input marginal trivial under IID; beliefs normalize; entropy rate $<$ log(vocab) (structure present).
- Rigor fixes vs prior experiment: (i) entropy-rate floor *computed*, not hardcoded; (ii) untrained baseline reported *per target*, not only joint.
- Notebook is torch/numpy only (no JAX) – avoids JAX/torch VRAM contention on one GPU; entropy-rate pre-check auto-flags degenerate (unpredictable) processes before training.

Run plan & next steps

Setup used: free-T4 Colab, `COLAB_FREE_T4=True` (100k steps, 10k analysis seqs, ∼40 min/run), Drive-persisted checkpoints per run. (Library tests: `uv run --extra dev --extra pytorch pytest -k "transducer or coarse"`.)

Done: iid→sns (validation), sns→sns and fractal2→fractal2 in both regimes + erasure cross-fit, and the degenerate-Fractal2 control.

Next:

1. Third process (mess3 → sns) to confirm the erasure pattern once more.
2. Full 1M-step runs for tighter numbers (on a non-free GPU).
3. Feedback / partially observable (`transducer_feedback_experiment.ipynb`): each agent's modified-operator MSP.

Questions / discussion